

On Performance of multi-hop assisted mMTC for DECT-2020 New Radio System

Sandhya Soni

ECE Engg.

The LNMIIT

Jaipur, India

18PEC002@lnmiit.ac.in

Rahul Makkar

ECE Engg.

The LNMIIT

Jaipur, India

rahul.makkar27@gmail.com

Tanmay Singhwi

ECE Engg.

The LNMIIT

Jaipur, India

17DEC006@lnmiit.ac.in

Divyang Rawal

Dept. of ECE

The LNMIIT

Jaipur, India

divyang.rawal@lnmiit.ac.in

Nikhil Sharma

Dept. of ECE

The LNMIIT

Jaipur, India

nikhil.sharma@lnmiit.ac.in

Laxmikant Minz

Dept. of EE

KAIST

Daejeon, South Korea

laxmikant.minz@gmail.com

Abstract—Digital enhanced cordless telecommunication new radio (DECT-2020 NR) is one of the candidate technology which fulfils the minimum requirements to serve as fifth-generation (5G) standardization. It accomplishes one of the performance requirements for massive machine-type communications (mMTC) by enabling a high connection density, i.e. up to 1-million devices/km². Reliable communication over such massive connections is achieved through a multi-hop relay system. This paper analyzes the link-level performance of DECT-2020 NR standards for mMTC using a multi-hop decode-and-forward (DF) relaying system for the outage probability and bit error rate (BER) performance metrics. Simulation results show the performance improvement in outage probability over various transmission rates, hops and signal-to-interference-plus-noise ratio (SINR). The numerical results corroborate with Monte-Carlo simulation results. Also, the link performance in terms of BER is compared for Turbo codes with puncturing.

Index Terms—5G, new radio (NR), digital enhanced cordless telecommunication (DECT), multi-hop, outage probability.

I. INTRODUCTION

Fifth-generation (5G) communication system is broadly classified into three features, namely enhanced mobile broadband (eMBB), ultra-reliable and low latency communication (uRLLC) and massive machine-type communication (mMTC) [1]. These areas serve multiple users or devices with reliable communication, high data rates, enhanced capacity, and low latency transmission capabilities in recent applications such as device-to-device (D2D), machine-to-machine (M2M), vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I) communication, etc. [2]. Apart from 3GPP technologies such as long term evolution (LTE) for machine type communication, new radio (NR) and narrowband - Internet of things (NB-IoT), digital enhanced cordless telecommunication (DECT) claims to satisfy the minimum technical criteria which address all of

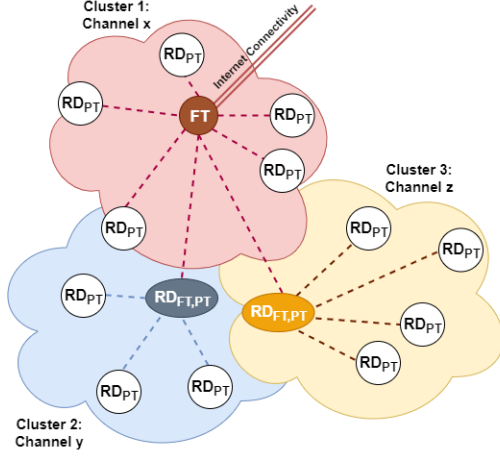
the needs stated above [4], [5]. Authors in [5] investigated the observations on the assessment of candidate technology component ETSI: DECT-2020 NR for mMTC and uRLLC usage scenario and illustrated the evaluation results.

The primary performance requirement for mMTC is to enable a massive connection up to 1-million devices/km², which is almost ten times higher than the existing LTE technology [4], [5]. DECT-2020 NR is an essential new standardization project aimed at meeting some of IMT-2020's criteria. It broadcasts in the 1880–1900 MHz range, with 10 frequency channels ranging from 1881.792 MHz to 1897.344 MHz with a bandgap of 1728kHz [6]. Similar to NR, DECT-2020 NR also provides a subcarrier scaling factor (μ) ranges {1, 2, 4, 8} results in subcarrier spacing {27, 54, 108, 216} kHz, respectively. Also, it provides the mesh network topology to connect multiple radio devices via fixed terminating (FT) point [7].

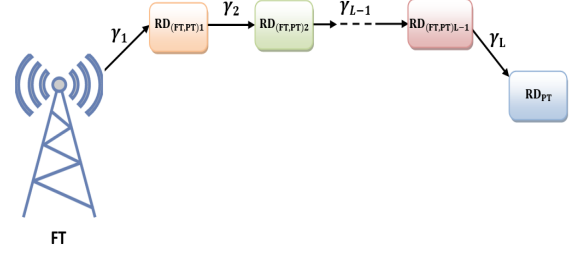
For reliable communication over a densely populated devices, the cooperative relaying technique results in the enhancement of quality of service (QoS) [8]. Outage probability (OP) is also one of the performance metrics to investigate the QoS. Literature shows that the analytical expression of outage probability has been derived over various fading environments [9], [10]. Authors in [11], [12] analyzed the performance in terms of outage probability for the decode-and-forward (DF) system over the Nakagami-m fading channel. Authors in [13], [14] investigated the performance of the multi-hop system over the generalized-K and fluctuating two-ray fading channels, respectively and showed the asymptotic outage probability. Also, authors in [14] showed the ASER performance of aforesaid multi-hop system. Bit error rate (BER) performance over NR specifications is investigated in [15] over multiple interfering devices.

Based on the above-described research of multi-hop DF system, we used the inherent merits of relaying communication for mMTC DECT-2020 NR environment and presented a

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(a) DECT-2020 NR system model with multiple FTs



(b) Considered multi-hop relaying system

Fig. 1: An illustration of DECT-2020 NR with multi-hop relaying system

composite system model. The contributions of the paper are summarised as follows.

- We investigate the outage probability of the multi-hop system for various transmission rates (R), the number of hops (L) and signal-to-interference-plus-noise ratio (SINR).
- The analytical expression of outage probability is investigated for aforesaid system over static and dynamic channel environment.
- BER performance of DECT-2020 NR specifications is also presented for various FFT points.

The remainder of the paper is summarized as follows. Section II presents the frame format of DECT-2020 NR specifications. Section III describes the system model for considered multi-hop relaying system. Section IV provides the analysis of outage probability for a multi-hop system with static and dynamic channels. Section V describes the numerical results, and the paper concludes in section VI.

II. FRAME STRUCTURE IN DECT

The frame structure of DECT-2020 NR is quite similar to NR. It provides the variable numerology factor (μ) that allows a different number of resource slots/frames with the subcarrier spacing variation $\Delta f = 27 \times \mu$ kHz. It uses an orthogonal frequency division multiplexing (OFDM) technique similar to LTE/NR and multiple OFDM symbols to construct frames, subframes, and subsequent slots. In DECT, 24 fixed slots consist of various OFDM symbols per slot, i.e. $10 \times \mu$. One radio frame consists of 24 slots for the time duration of 10ms. Further based on μ , each slot divided into subslots with varying OFDM symbols. With the variation in $\mu = \{1, 2, 4, 8\}$, the number of subslots and OFDM symbols in each slot varies as $2 \times \mu$ and $10 \times \mu$, respectively [4], [6]. Table I shows the DECT-2020 NR frame format for various numerology [7] and simulation parameters.

TABLE I: DECT Frame Format

Parameters	DECT
Maximum Bandwidth	10 MHz
Symbol duration	10 ms
Numerology factor (μ)	$\{1, 2, 4, 8\}$
Slots per frame	Fixed
Subcarrier spacing (Δf)	$27 \times \mu$ kHz
OFDM symbols/slot	$10 \times \mu$
FFT points(*)	$\{64, 128, 256, 1024\}$
Useful subcarriers(*)	$\{56, 112, 224, 896\}$
Cyclic prefix(*)	$\{8, 16, 32, 128\}$
Channel coding	Turbo Codes (CTC-1/3)
Note: (*) denotes the considered values used in simulation.	

III. SYSTEM MODEL

Fig. 1a shows the system operation with multiple fixed transmission points (FT). Here, RD_{PT} is the terminology used to represent the receiving devices and $RD_{FT,PT}$ is used for a radio devices acting as intermediate relays that re-transmit the decoded data [4].

With reference to DECT-2020 NR system model, a single link multi-hop relaying system as shown in Fig. 1b is considered. Here, the fixed source (FT) transmits the modulated information data to the destination (RD_{PT}) through $(L - 1)$ intermediate relays. All the $(L - 1)$ intermediate relays are expressed as $RD_1, RD_2, \dots, RD_{L-1}$, respectively and they utilize the decode-and-forward (DF) cooperative scheme. The average signal-to-noise ratio (SNR) of the l^{th} hop is denoted by γ_l , where $l \in \{1, 2, \dots, L\}$. Let the transmitted signal from FT in time-domain is x and the decoded signal at i^{th} relay node is $\hat{x}_i$, where $i \in \{1, 2, \dots, (L - 1)\}$. Considering the independent and identically distributed (i.i.d.) Rayleigh fading coefficient distributed as $\sim \mathcal{CN}(0, 1)$, the received signal at l^{th} node is given as

$$y_l = \sqrt{P} h_l \hat{x}_i + n_l, \quad (1)$$

where P is the transmitting power, $\hat{x}_i$ is the decoded symbol at $i = (l-1)^{\text{th}}$ relay with $\hat{x}_0 = x$, n_l denotes the additive white Gaussian noise (AWGN) at the l^{th} node $\sim \mathcal{CN}(0, N_0)$. Without any loss of generality, the instantaneous SNR (γ_l) over each hop is considered same for all the links and given as $\gamma_l = |h_l|^2 \gamma$, where $\gamma = \frac{E_b}{N_0}$ is the average SNR with average bit energy (E_b).

DECT employs the OFDM transmission, in which modulated symbols are allocated to orthogonal resource subcarriers utilizing N -point FFT/filter banks. In frequency-domain, the received signal for m^{th} subcarrier is given by

$$Y_l(m) = H_l(m) \sqrt{\frac{P}{N}} \hat{x}_i(m) + N_l(m), \quad (2)$$

where H_l is the N -point frequency representation of L_{ch} Rayleigh faded channel coefficients and $N_l \sim \mathcal{CN}(0, N_0)$ is the receiver noise at l^{th} hop, respectively.

IV. OUTAGE PROBABILITY

This section presents the performance analysis of considered multi-hop relaying system in terms of outage probability and BER for static and dynamic channel environments.

The outage event of a multi-hop relaying system occurs when at least a single-hop has its SNR below the threshold SNR (γ_{th}). Thus, the outage probability (P_{out}) of the L -hops over a given transmission rate (R) in bits per second (bps) can be defined as

$$P_{out} = 1 - \prod_{l=1}^L P_r(\gamma_l > \gamma_{th}), \quad (3)$$

where $P_r(\cdot)$ is the probability operator and $\gamma_{th} = 2^{(LR)} - 1$. After simplifying the above mathematical expression, the P_{out} for multi-hop system can be expressed as [14]

$$\begin{aligned} P_{out} &= 1 - \prod_{l=1}^L [1 - F_{\gamma_l}(\gamma_{th})], \\ &= \sum_{l=1}^L F_{\gamma_l}(\gamma_{th}) - \sum_{\substack{k=1, l=1 \\ k \neq l}}^L F_{\gamma_k}(\gamma_{th}) F_{\gamma_l}(\gamma_{th}) \cdots \\ &\quad + (-1)^L \prod_{l=1}^L F_{\gamma_l}(\gamma_{th}), \end{aligned} \quad (4)$$

where $F_{\gamma_l}(\cdot)$ denotes the cumulative distribution function (CDF) of γ_l .

A. Static Channel

For a static channel over Rayleigh fading distribution with the given γ_{th} , the $F_{\gamma}(\cdot)$ can be given as

$$F_{\gamma}(\cdot) = 1 - \exp\left(-\frac{\gamma_{th}}{\gamma}\right). \quad (5)$$

B. Dynamic/Frequency-selective Channel

For a dynamic/frequency-selective channel over Rayleigh fading distribution with the given γ_{th} and channel taps (L_{ch}), the $F_{\gamma}(\cdot)$ can be given as

$$F_{\gamma}(\cdot) = 1 - \exp\left(-\frac{\gamma_{th}}{\Gamma}\right), \quad (6)$$

where $\Gamma = \left(\frac{L_{ch}}{N}\right) \gamma$ is a factor used for converting the multipath fading scenario to flat fading.

V. NUMERICAL RESULTS AND DISCUSSION

In this section, the simulation results in terms of outage probability and BER of a multi-hop system is presented. For each hop, we assume the channel to be i.i.d. Rayleigh fading, for which the information is known at each node. For outage analysis, various transmission rates $R = \{1, 2, 3, 4, 5\}$, number of hops $L = \{2, 3, 4, 20\}$ and SINR $\{5, 10, 15\}$ dBm are considered. Based on the different DECT-2020 NR configurations over various FFT points, we chose $L_{ch}=5$ channel taps for multipath fading. For BER analysis, we consider $\mu=1$ with 10 OFDM symbols per slot. BPSK modulated symbols are transmitted over aforesaid frame considerations. Simulation results consist various number of FFT points with $N = \{64, 128, 256, 1024\}$.

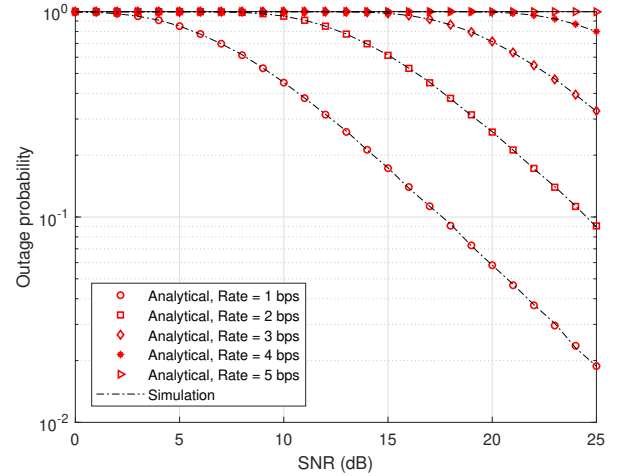


Fig. 2: Outage performance of dual-hop system with transmission rate $R = \{1, 2, 3, 4, 5\}$ bps

Fig. 2 shows the outage performance of the dual-hop system with different rates $R = \{1, 2, 3, 4, 5\}$ bps. It is observed that for a considered dual-hop system, the outage probability increases with an increase in the rate, justifying as the threshold increases with the transmission rate. Fig. 3 depicts the performance of the system for a multi-hop system with $L = \{3, 4, 20\}$ over a constant transmission rate $R = 1$ bps. It is observed that the outage probability of the system increases in proportion with the number of hops.

To know the impact of outage performance of the considered system over varying SINR, we also simulate and show the result in Fig. 4 considering SINR $\{5, 10, 15\}$ dBm

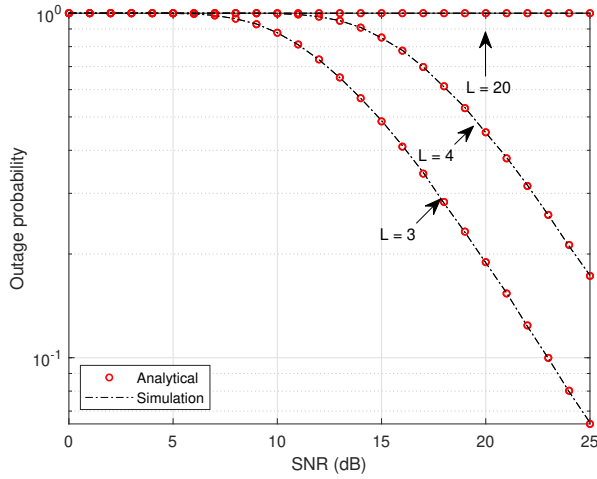


Fig. 3: Outage performance of multi-hop system with hops $L = \{3, 4, 20\}$ and transmission rate $R = 1$ bps

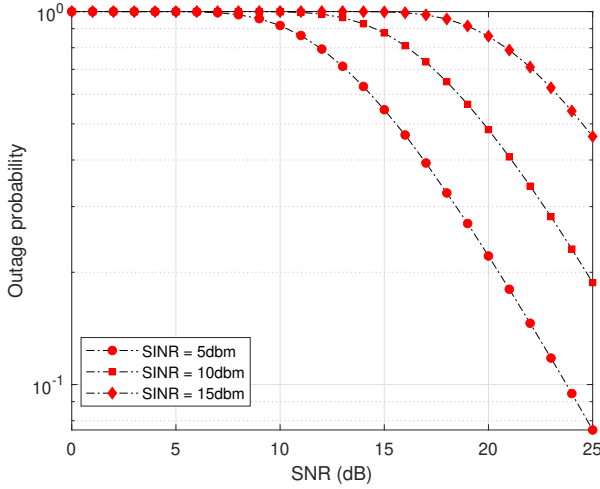


Fig. 4: Outage performance of dual-hop system with various SINR $\{5, 10, 15\}$ dBm

for a dual-hop system with $R = 1$ bps. It is observed that increasing the SINR (in dBm) results in more unsatisfactory outage performance, which validates the simulated results.

Fig. 5 describes the comparative BER performance of single-hop coded DECT-2020 NR transmission with different FFT points over BPSK transmission. For simulations, $\{56, 112, 224, 896\}$ useful subcarriers is transmitted for $\{64, 128, 256, 1024\}$ FFT points, respectively. Also, a code rate of $1/3$ is considered for turbo coders without puncturing bits. It is observed that for smaller FFT points, the performance is improved as the subcarrier power reduces with increase in the number of FFT points.

Fig. 6 explains the comparative analysis of single-hop uncoded and coded transmission with a various code rate of $1/3$ (without puncturing bits) and $9/10$ (with puncturing bits) over DECT-2020 NR specifications. The impact of channel coding

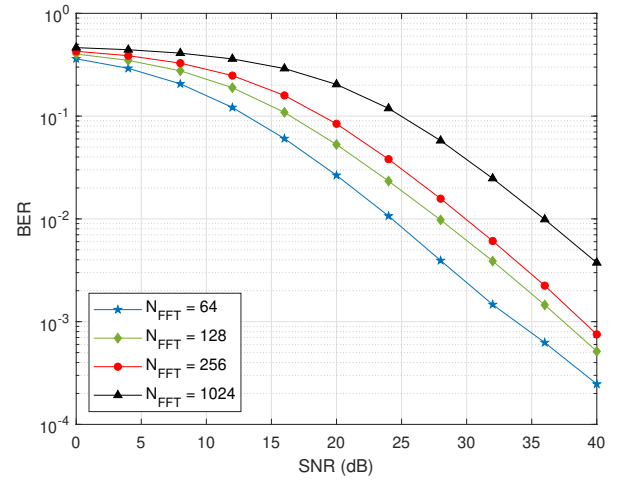


Fig. 5: BER performance of single-hop coded system with various FFT points

enhances the BER performance over uncoded transmission. For an error floor of 10^{-2} , there is a significant improvement in SNR of almost 3-5 dB.

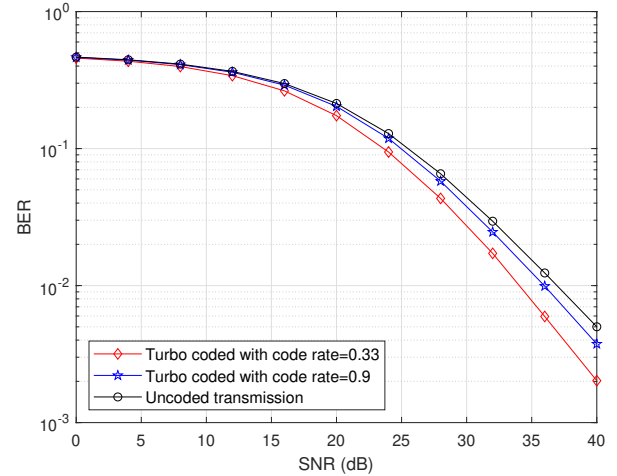


Fig. 6: BER performance comparison of coded and uncoded system for various code rate

VI. CONCLUSION

This paper provides the simulation results of DECT-2020 NR specifications for a multi-hop relaying system over Rayleigh fading channel. The outage performance of the proposed system deteriorates with increasing number of hops, transmission rate and SINR for mMTC DECT-2020 NR environment. Also, the BER performance of coded system improves with less puncturing, i.e. 2-3 dB improvement.

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